Topic in Data Mining: Geospatial Health Data

INFO 523 Final Project

Carmen Soto

Data Mining Topic: Geospatial health data

Final project – INFO52 PRESENTATION:

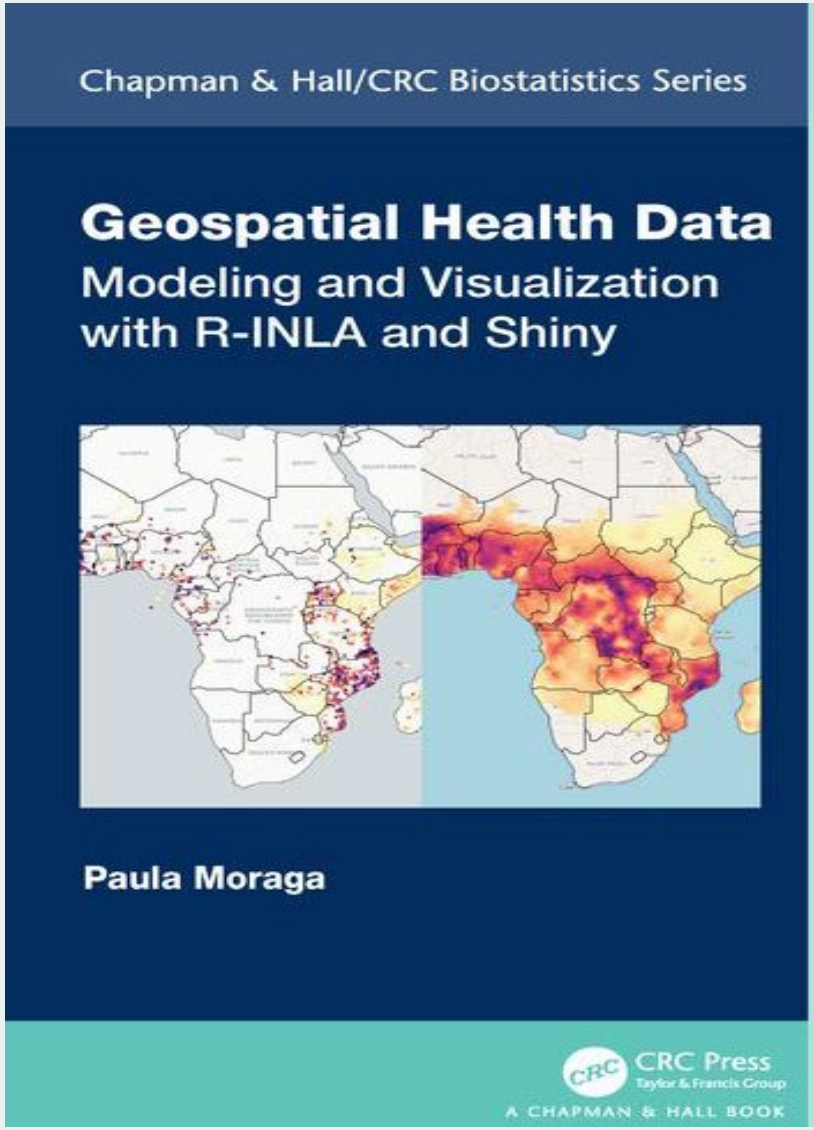
- 1. Background information and purpose of the model/technique/method
- 2. Details on how this topic relates to any of the topics in the class
- 3. Real-world applications
- 4. A worked example in R/

GitHub repo: [carmssoto/INFO523_Final_Project_CarmenSoto: Carmen Soto Final Project]

5. Conclusions, recommendations, and future work

Background information and purpose of the model/technique/method : **Geospatial health data**

resource link: [Welcome | Geospatial Health Data: Modeling and Visualization with R-INLA and Shiny]



How is Geospatial health data, data mining?

Geospatial health data mining involves the use of spatial and spatio-temporal statistical methods to analyze georeferenced health data in R.

<u>The process involves:</u>

- Data importing
- Manipulation
- Modeling and visualization
- Presentation of results: creating R markdown reports, interactive dashboards, Shiny web applications

Ensuring results are clear and accessible for stakeholders.

Geospatial health data: Background information and purpose

1.1 Geospatial health data

Health data provides information to identify public health problems and respond appropriately when they occur. This information is crucial to prevent and control a variety of health conditions such as infectious diseases, non-communicable diseases, injuries, and health-related behaviors. The process of analysis and interpretation of health data encompasses a broad variety of system designs, analytic methods, modes of presentation, and interpretive uses (Lee et al. 2010).

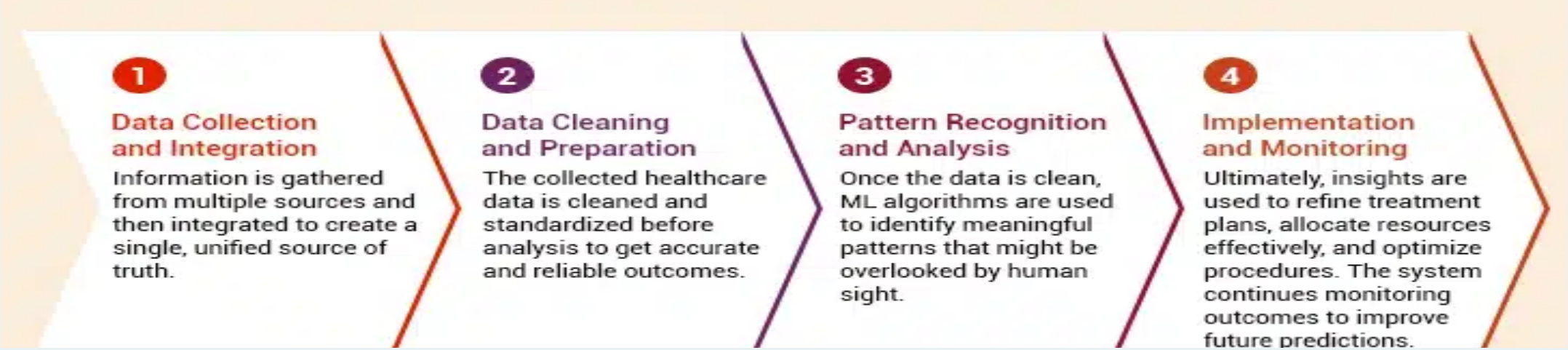
1.2 Disease mapping

The mapping of disease risk has a long history in public health surveillance. Disease maps provide a rapid visual summary of spatial information and allow the identification of patterns that may be missed in tabular presentations (Elliott and Wartenberg 2004).

1.3 Communication of results

It is important to note that the goal of health surveillance is not merely to collect data for analysis, but to guide public health policy and action to control and prevent diseases.

Moraga, P. (2016). Chapter 1 Geospatial health | Geospatial Health Data: Modeling and Visualization with R-INLA and Shiny. Paulamoraga.com. https://www.paulamoraga.com/book-geospatial/sec-intro.html



How Geospatial health data is used in Health Disparities Research

Geospatial health data mining is essential for understanding disease burden, geographics, and patterns. With the goal to identify risk factors and measuring health inequalities.

- Method is used to identify geographic Health Disparities in minority populations, can be used to implement policy changes, health programs, resource allocations.

R packages useful to create static and interactive maps including

ggplot2 (Wickham et al. 2024)
leaflet (Cheng et al. 2024)
mapview (Appelhans et al. 2023)
tmap (Tennekes 2023).

8 Geostatistical data

Geostatistical data are measurements about a spatially continuous phenomenon that have been collected at particular sites. This type of data may represent, for example, the disease risk measured using surveys at specific villages, the level of a pollutant recorded at several monitoring stations, and the density of mosquitoes responsible for disease transmission measured using traps placed at different locations (Waller and Gotway 2004). Suppose $Z(\boldsymbol{s}_1), \ldots, Z(\boldsymbol{s}_n)$ are observations of a spatial variable Z at locations $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n$. In many situations, geostatistical data are assumed to be a partial realization of a random process

$$\{Z(\boldsymbol{s}) : \boldsymbol{s} \in D \subset \mathbb{R}^2\},$$

where D is a fixed subset of $\mathbb{R}^2$ and the spatial index $\boldsymbol{s}$ varies continuously throughout D. Many times, the process $Z(\cdot)$ can only be observed at a finite set of locations for practical reasons. Based upon this partial realization, we seek to infer the characteristics of the spatial process that gives rise to the data observed such as the mean and variability of the process. These characteristics are useful for the prediction of the process at unobserved locations and the construction of a spatially continuous surface of the variable of study.

Moraga, P. (2016). Chapter 1 Geospatial health | Geospatial Health Data: Modeling and Visualization with R-INLA and Shiny. Paulamoraga.com. https://www.paulamoraga.com/book-geospatial/sec-intro.html

Educational Resource:

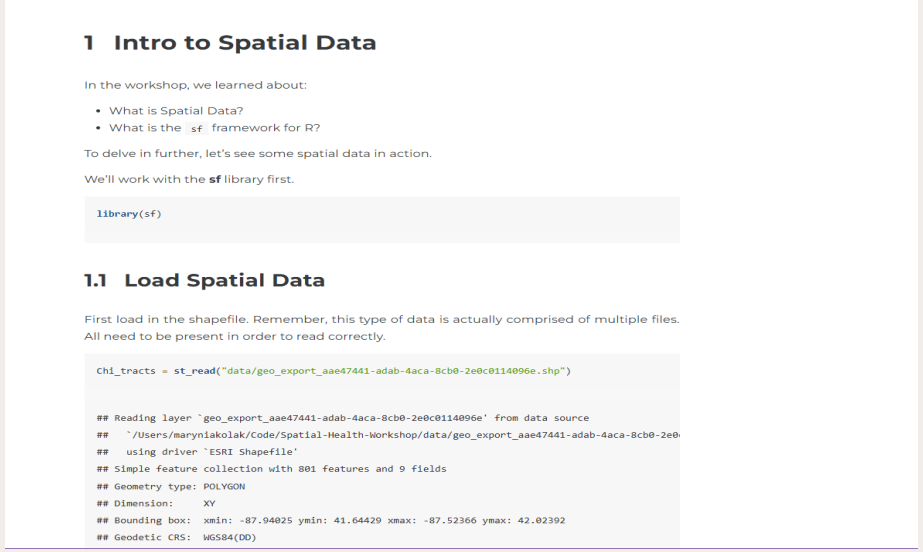
Intro to Spatial Analysis in R for Health & Geospatial Data Analysis with R

1) Workbook: Github Link: Intro to Spatial Analysis in R for Health

https://healthyregions.github.io/Spatial-Health-Workshop/index.html

Intro to Spatial Analysis in R for Health. (2025, June 18). Github.io.

This educational resource is led by Marynia Kolak, Director of the Healthy Regions & Policies Lab (HEROP), in this guide you learn how Spatial Analysis us used to explore how **social**, spatial, racial, built environment, and structural **drivers** influence **health**. (2026). Healthyregions.org. https://healthyregions.org/



1 Intro to Spatial Data

In the workshop, we learned about:

- What is Spatial Data?
- What is the `sf` framework for R?

To delve in further, let's see some spatial data in action.

We'll work with the **sf** library first.

```
library(sf)
```

1.1 Load Spatial Data

First load in the shapefile. Remember, this type of data is actually comprised of multiple files. All need to be present in order to read correctly.

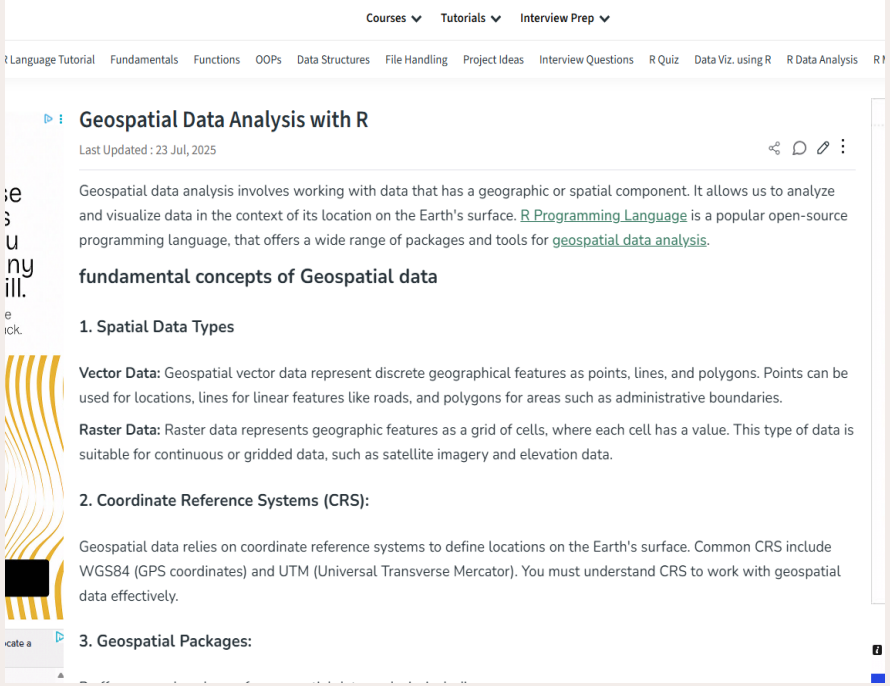
```
Chi_tracts = st_read("data/geo_export_aae47441-adab-4aca-8cb0-2e0c0114096e.shp")

## Reading layer `geo_export_aae47441-adab-4aca-8cb0-2e0c0114096e' from data source
##   `/Users/maryniakolak/Code/Spatial-Health-Workshop/data/geo_export_aae47441-adab-4aca-8cb0-2e0
##   using driver `ESRI Shapefile'
## Simple feature collection with 801 features and 9 fields
## Geometry type: POLYGON
## Dimension:     XY
## Bounding box:  xmin: -87.94025 ymin: 41.64429 xmax: -87.52366 ymax: 42.02392
## Geodetic CRS:  WGS84(DD)
```

2) Guide: Geospatial Data Analysis with R

Geospatial Data Analysis with R – GeeksforGeeks

https://www.geeksforgeeks.org/r-language/geospatial-data-analysis-with-r/

GeeksforGeeks. (2023, October 16). *Geospatial Data Analysis with R*. GeeksforGeeks. https://www.geeksforgeeks.org/r-language/geospatial-data-analysis-with-r/



Courses Tutorials Interview Prep

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Geospatial Data Analysis with R

Last Updated : 23 Jul, 2025

Geospatial data analysis involves working with data that has a geographic or spatial component. It allows us to analyze and visualize data in the context of its location on the Earth's surface. R Programming Language is a popular open-source programming language, that offers a wide range of packages and tools for geospatial data analysis.

fundamental concepts of Geospatial data

1. Spatial Data Types

Vector Data: Geospatial vector data represent discrete geographical features as points, lines, and polygons. Points can be used for locations, lines for linear features like roads, and polygons for areas such as administrative boundaries.

Raster Data: Raster data represents geographic features as a grid of cells, where each cell has a value. This type of data is suitable for continuous or gridded data, such as satellite imagery and elevation data.

2. Coordinate Reference Systems (CRS):

Geospatial data relies on coordinate reference systems to define locations on the Earth's surface. Common CRS include WGS84 (GPS coordinates) and UTM (Universal Transverse Mercator). You must understand CRS to work with geospatial data effectively.

3. Geospatial Packages:

2. Details on how this topic relates to any of the topics in the class

How Geospatial Health is Data Mining in Health Care

How Geospatial Analysis is Data Mining

#2 How this topic relates to topics in our INFO523 Class

1)Introductory Lecture: Geospatial aligns with the biological and medical data analysis examples of data mining mentioned in introduction lectures

2)Factors and Categorical Data In Lectures 2 and 5 we were introduced to Factors as Variables for categorical data

3) Week 4: Ggplot: In Lecture 8 we learned about ggplot and using dplyr. Geospatial data expands on the powerful ways to display using maps & heat maps to display findings.

#3 Real-world applications

1. World Health Organization:

WHO GIS Centre for Health:
Geospatial technology in the form of
Geographic Information Systems (GIS) enables spatial representation of data to support better public health planning and

- ➢ Geolocated Health Facilities Data initiative

Geolocated Health Facilities Data initiative. (n.d.). Www.who.int. https://www.who.int/data/GIS/GHFD

Geolocated Health Facilities Data Initiative

The Geolocated Health Facilities Data (GHFD) initiative is a **collaborative effort** that provides support to countries who need assistance updating, geolocating, digitizing, and/or openly sharing the health facility master list (HFML) for their country. As a **global public good**, the final product will be the world's first central and accessible public directory of health facility names, locations and types.

GOAL

Ministries of Health (MOH) are in the position to maintain, regularly update, and use the health facility master list of their respective country. Elements of these lists are shared in a global directory, which will serve as a global public good.

MISSION

Strengthen the technical capacity of MOHs across levels to ensure the availability, quality, accessibility, and use of HFML data by building on what already exists to avoid duplication of efforts and address public health priorities.

Health Facility Master Lists

One HFML per country that is digitized, regularly updated, and made accessible by the MOH.

+

Country Capacity

MOH capacity to regularly maintain, update, use, and publicly share their HFML.

+

Global Directory

Publicly available directory of digitized health facility information sourced from country HFMLs.

VALUE

- Increase country-level capacity to maintain a regularly updated HFML that is seen as the singular source of health facility data in a country
- Make geolocated health facility data accessible
- Leverage and use data rather than simply collect it
- Strengthen health systems and delivery
- Increase data interoperability while avoiding duplication of efforts
- Advance progress toward the World Health Organization's Triple Billion targets and Sustainable Development Goals

World Health Organization

Real world Applications & Worked Example In R

Medical Data Analysis & Targeted Intervention: Research topic- Maternal Sepsis

Maternal sepsis is the 3rd most common cause of maternal mortality

Definition: Maternal sepsis is a life-threatening condition that arises when the body's response to infection causes injury to its own tissue and organs during pregnancy, childbirth, post-abortion or the post partum period. (World Health Organization, 2021)

Real world application: NYS DATA ANALYSIS- END SEPSIS

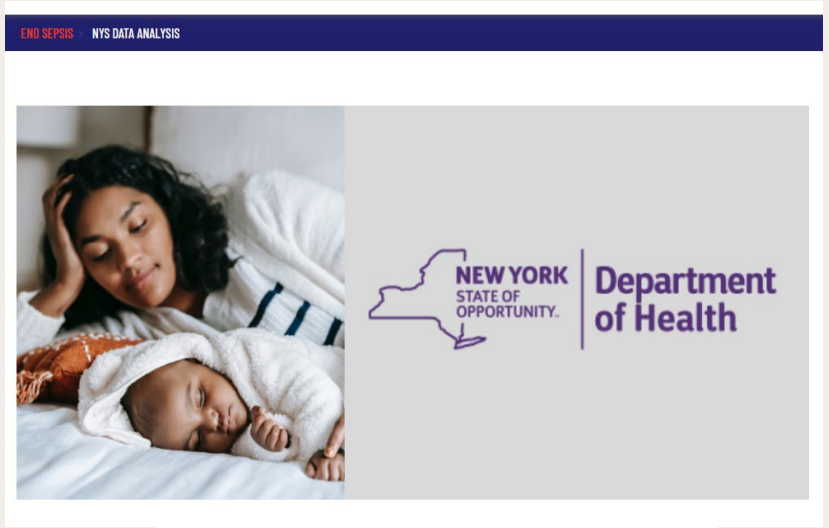
In collaboration with END SEPSIS, the New York State Department of Health created a dataset that documented cases of maternal sepsis in New York State from 2016 to 2018.

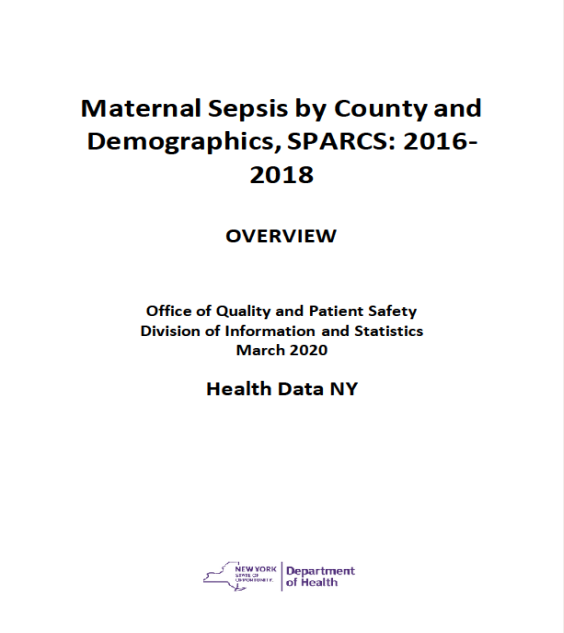
By merging data from the Statewide Planning and Research Cooperative System (SPARCS) and Vital Statistics datasets, the NYSDOH was able to identify potential at-risk populations by analyzing associations between hospitalizations for maternal sepsis and hospital and patient characteristics.

The study analyzed both live births and stillbirths and investigated the following factors: Demographics, Hospital Characteristics Comorbidities, Delivery/Obstetric Characteristics

Link: NYS Data Analysis - End Sepsishttps://www.endsepsis.org/nys-maternal-sepsis-data-analysis/

(NYS Data Analysis - End Sepsis, 2023)





Maternal Sepsis by County and Demographics, SPARCS: 2016-2018

OVERVIEW

Office of Quality and Patient Safety
Division of Information and Statistics
March 2020

Health Data NY

A worked example in R/python using real or simulated data: SPARCS: 2016-2018

"Maternal Sepsis by County and Demographics, SPARCS: 2016-2018"

Download link: https://health.data.ny.gov/api/views/f7ik-e57z/rows.csv?accessType=DOWNLOAD

(*State of New York*, 2018)

Goal is to analyze:

Pattern discovery of high sepsis rates among specific minority group per county locations with high rates reported by regions.

Link to my GitHub repo: carmssoto/INFO523_Final_Project_CarmenSoto: Carmen Soto Final Project

How I used Data Mining in my worked example in R

1. Discovery of Patterns
2. Outlier analysis

By mapping country- level data it allowed me to identify Cortland county had the highest recorded sepsis rate for Hispanic patients. (4,166 per 100k)

3) Transformed Maternal Sepsis CSV into a visual "hotspot map"

R CODE NOTES:

#**1ST STEP: Data Quality and clean process

#**2ND STEP: Smaller data frames Data Cleaning

#**STEP 3: HISPANIC SEPSIS RATES

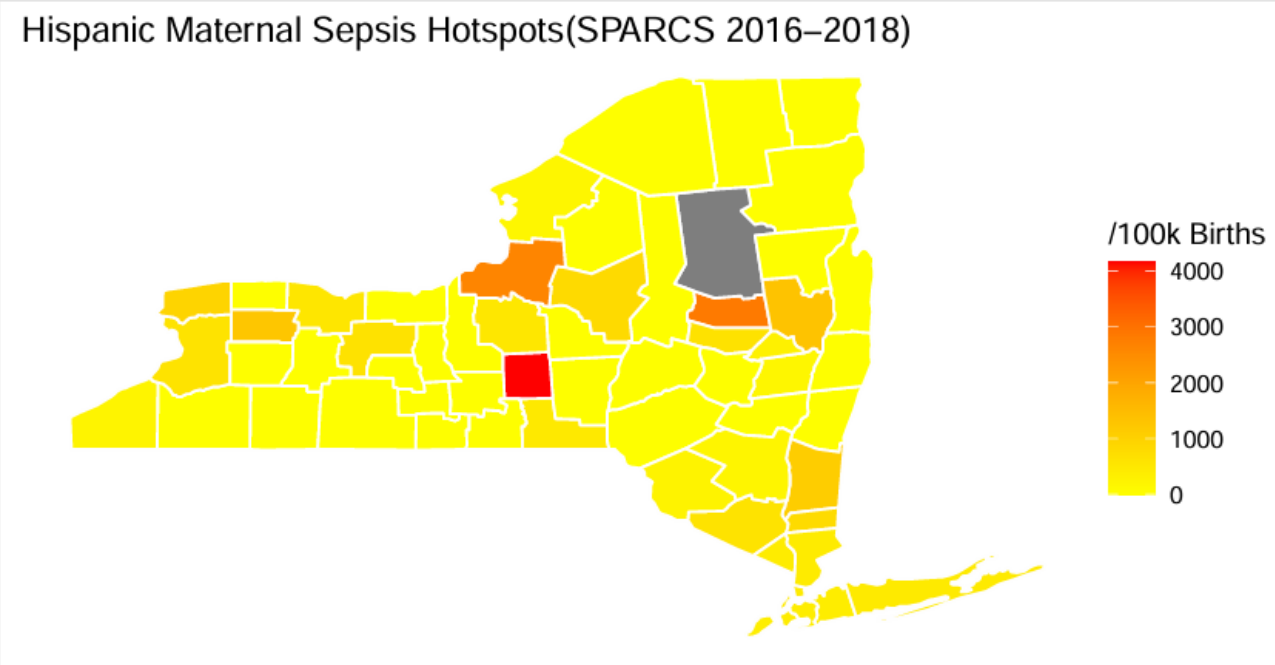
#**STEP 4: Geospatial Worked Example

```
#**1ST STEP Data Quality and clean process
#Read through the Sources the data dictionary and Overview documents
colnames(sepsis_raw)
#All Sepsis counts presented in the data set is used as the numerator for calculation of sepsis rates
#"Any Sepsis Events During Pregnancy (N)" - The number of all maternal sepsis events
#check total Patient Sepsis Cases
class(sepsis_raw$`Any Sepsis Events Peripartum (N)`)#[1] "character"
#change to numeric
sepsis_raw <- sepsis_raw %>%
  mutate(`Any Sepsis Events Peripartum (N)` =as.numeric(`Any Sepsis Events Peripartum (N)`))
class(sepsis_raw$`Any Sepsis Events Peripartum (N)`)#"numeric"
#Check which columns I will be working with:
#Patient county,Demographics,#of Live Births,# total Sepsis Cases
unique(sepsis_raw$`Patient County`)
unique(sepsis_raw$`Demographic Strata`)
unique(sepsis_raw$`Year(s) of Live Birth`)
unique(sepsis_raw$`Live Births\n(N)`)#"2016-2018"
unique(sepsis_raw$`Any Sepsis Events Peripartum (N)`)
sum(is.na(sepsis_raw$`Any Sepsis Events Peripartum (N)`))#13 missing values

#**2ND STEP Smaller data frames Data Cleaning

#SMALL RAW DATA
#I first make my small data frame with my selected columns
#rename my columns to clean up my code

sepsis_small<-sepsis_raw%>%
  select(`Patient County`,`Demographic Strata`,`Live Births\n(N)`,`Any Sepsis Events Peripartum (N)`)%
  rename(
    Patient County= `Patient County`
```

Real Worked Example In R:

Geospatial Analysis- Mapping Disparities Across Regions To Guide Targeted Interventions

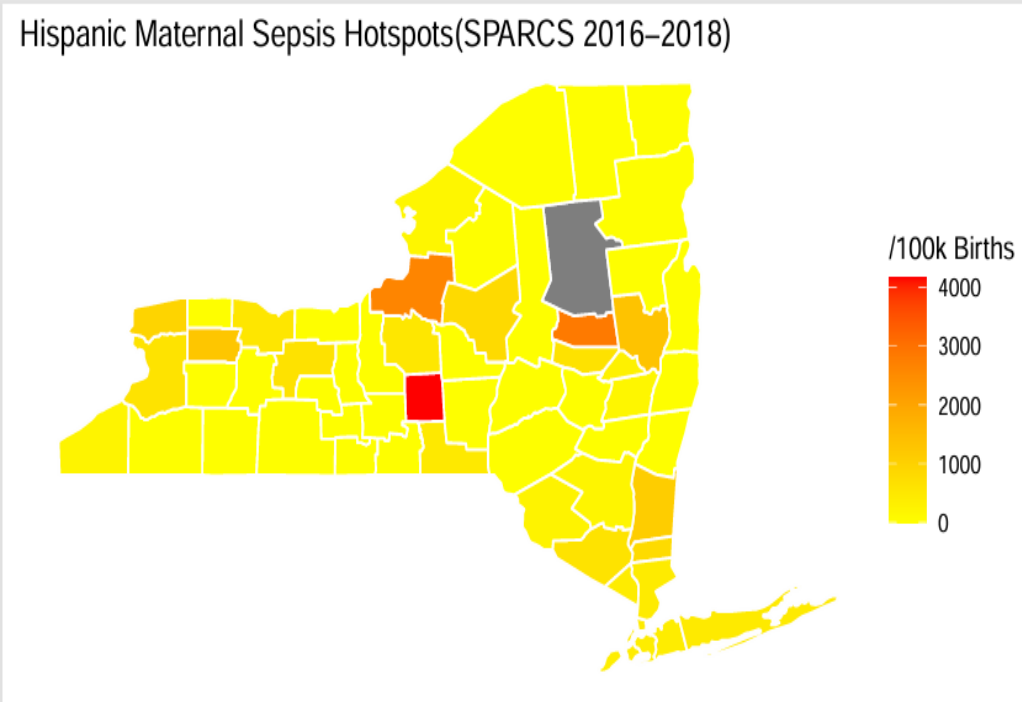
- 62 counties in New York State only 61 counties reported sepsis data during the years of 2016-2018
- Cortland County has the highest Maternal Sepsis Rate among Hispanic patients reported

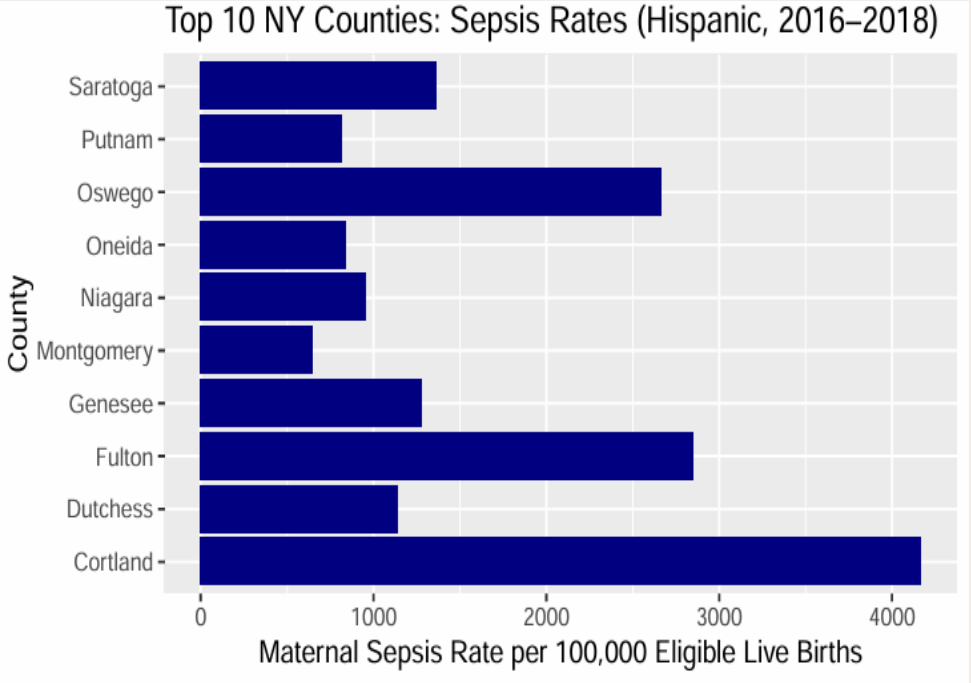


AI USE DISCLOSURE

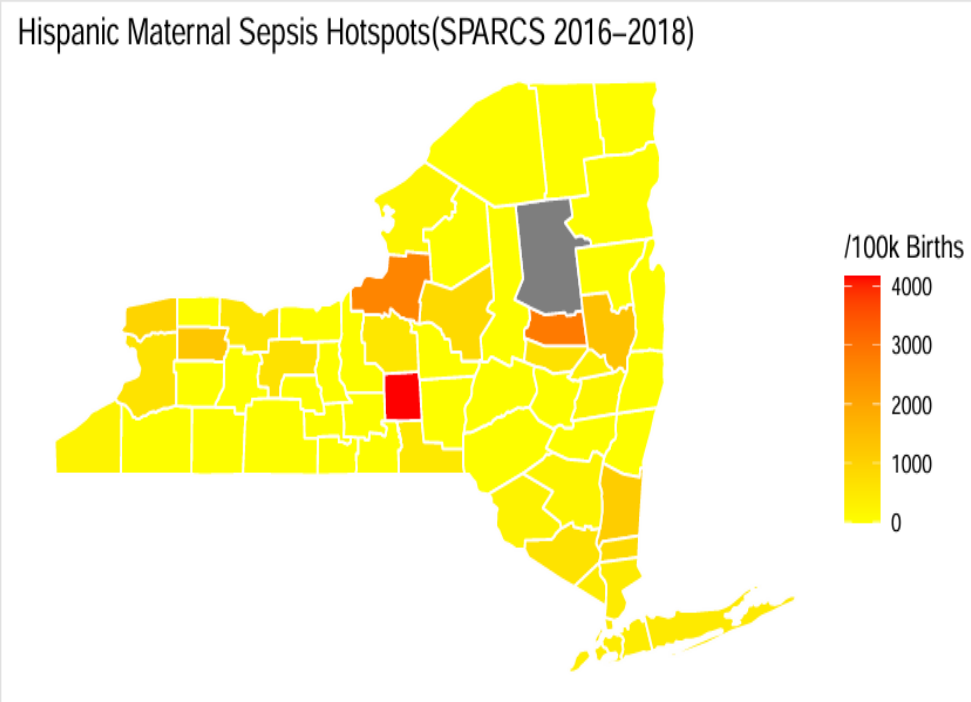
Portions of the R code and workflow development were assisted by ChatGPT (OpenAI), an AI language model. The tool was used to help troubleshoot code, suggest data manipulation strategies, and guide geospatial mapping techniques. All code was critically evaluated, modified as necessary, and executed by the author. The author takes full responsibility for the accuracy, integrity, and interpretation of the results.

Bar plot of top 10 NY counties by Hispanic maternal sepsis rates.





Regions of New York State



Conclusions, recommendations, and future work: real world applications END SEPSIS

The Maternal Sepsis Initiative: in Collaboration with END SEPSIS and the New York state department of Health was awarded funding through the Department of Health and Human Services Barda Drive a program developed to address maternal sepsis and provide patient facing educational materials and resources for prevention. Along with provider facing educational materials and intervention planning.

A continuum of care was developed to identify opportunities for intervention to help prevent, identify and rapidly treat maternal sepsis. This work is being used to help clinicians, nurses, hospital administrators, birth workers and others engaged in the care of pregnant and postpartum women better serve their patients. It is also a powerful tool for remedying the severe inequities in maternal care in New York State and beyond.

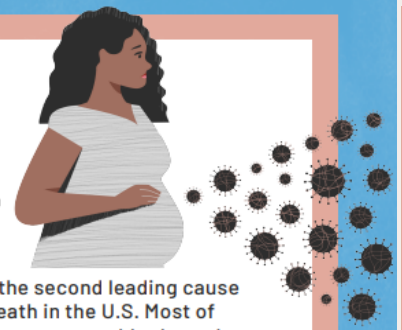
Link: https://www.endsepsis.org/maternal-sepsis-initiative/

Findings of note include:

- Women under 20 years old at the time of birth had significantly higher odds of developing maternal sepsis than women ages 20-35.
- All women with an education level below a four-year college degree had significantly higher odds of developing maternal sepsis across all windows. This is more pronounced among women with less than a high school education.
- Black, Hispanic, and Asian women had significantly higher odds of developing maternal sepsis compared to White women.
    - Hispanic women are more than twice as likely as White women to develop maternal sepsis
    - Black women are nearly twice as likely as White women to develop maternal sepsis
    - Asian Women are almost 1.5 times as likely as White women to develop maternal sepsis
- Women with C-sections had significantly increased odds of sepsis during delivery and postpartum.

(Maternal Sepsis Initiative, n.d.)

Maternal Sepsis Provider Fact Sheet



The United States has the highest rate of maternal mortality of any wealthy country–and those rates have risen over the past 20 years.

As defined by the World Health Organization, sepsis is "A life-threatening condition defined as organ dysfunction resulting from infection during pregnancy, childbirth, post-abortion, or postpartum period."

Maternal Sepsis, which is sepsis that occurs during pregnancy or in the 42 days following delivery or other obstetric event, is now the second leading cause of maternal death in the U.S. Most of these deaths are preventable through early identification and treatment of the underlying infection.

Through new analysis of administrative discharge data and birth record data from 2016-2018, the New York State Department of Health (NYS DOH) has identified factors that put some patients at increased risk of developing maternal sepsis. Interventions that could improve outcomes for pregnant and postpartum women have also been identified.

Demographic Risk Factors

- **Women of color** have significantly higher odds of developing maternal sepsis compared to white women:
  - **Hispanic women** have **2.1 times** the odds of developing maternal sepsis compared to White women
  - **Black women** have **1.8 times** the odds of developing maternal sepsis compared to White women
  - **Asian women** have **1.4 times** the odds of developing maternal sepsis compared to White women
- Women **less than 20 years old** at time of birth are twice as likely to develop maternal sepsis compared to women ages 20-35.
- Women with an education level **below a Bachelor's Degree** experience significantly higher odds of sepsis. This increase is most pronounced among women with less than high school education.

Obstetric Risk Factors and Maternal Morbidities

The vast majority of maternal sepsis cases occur within two weeks of delivery (or other obstetric event) and are associated with complications during childbirth.

- The most common risk factor is Cesarean delivery. Women who deliver via C-section are three times as likely to develop maternal sepsis and those who deliver vaginally.

Other Prevalent Maternal Morbidities/Obstetric Risk Factors identified using NYS DOH data include:

- **Induction of labor**
- **Nulliparity (never having given birth)**
- **Premature rupture of membranes**
- **Preterm delivery**
- **Postpartum hemorrhage**
- **Blood products transfusion**
- **Multiple gestation**

ENDSEPSIS

Next Page ▸

Patients with sepsis may complain of the following symptoms:

- **Fever and chills and generally feeling unwell**
- **Dizziness and confusion**
- **Back or abdominal pain**
- **Foul-smelling vaginal discharge**
- **Feeling like their heart is racing**
- **Chesty cough**
- **Pain/burning on passing urine or not passing urine in a day**

Comorbidities

The vast majority of comorbidities analyzed in the report were associated with significantly increased risk of maternal sepsis.

Prevalent comorbidities with an opportunity for intervention:

- **Illegal drug use**
- **Tobacco use**
- **Obesity**
- **Deficiency anemia**
- **Asthma**
- **Depression**
- **Sickle cell disease**
- **History of sepsis (within 1yr of pregnancy)**

The clinical signs of maternal sepsis:

Any of these abnormal criteria, sustained for >20 minutes:

- **Systolic BP (mmHg) <90 or >160**
- **Diastolic BP (mmHg) >100**
- **Heart Rate <50 or >120**
- **Respiratory Rate <10 or >24**
- **O2 Sat on room air; % <95**
- **Oliguria, mL/hr x 2hrs <35**
- **Temperature <36 C or >38 C**
- **WBC <4,000 or >15,000**

Maternal agitation, confusion, or unresponsiveness; patient with hypertension reporting a non-remitting headache or shortness of breath.

The complete maternal sepsis datasets are available here:
Maternal Sepsis by County and Demographics, SPARCS: 2016-2018
Maternal Sepsis by Select Risk Factors, SPARCS: 2016-2018

Possible areas of intervention

Increased vigilance when caring for those women identified as being at increased risk of developing sepsis (i.e. young women and women of color) and those who experience complications during delivery, including C-section and induced delivery. (A maternal early warning system is recommended by The American College of Obstetricians and Gynecologists District II: Safe Motherhood Initiative Maternal Early Warning System.)

Appropriate discharge planning for those women identified as being at increased risk for maternal sepsis as the vast majority of maternal sepsis cases occur in the early postpartum period. This includes listening carefully to the concerns of new mothers who may articulate symptoms that indicate possible sepsis, providing close follow-up after discharge and ensuring that patients and their families are familiar with the signs of maternal sepsis.

Targeting emergency department providers with education on maternal sepsis for women who visit during pregnancy. More than half of all maternal sepsis cases were preceded by an ED or inpatient hospital visit and most (excluding obstetric events) were ED visits. (A maternal early warning system is recommended by The American College of Obstetricians and Gynecologists District II: Safe Motherhood Initiative Maternal Early Warning System.)

ENDSEPSIS
The Legacy of Rory Staunton
endsepsis.org

Sources:
Center for Disease Control and Prevention. 2021. Pregnancy Mortality Surveillance System.
World Health Organization. 2021. Maternal Sepsis.
ACOG Safe Motherhood Initiative. 2021.

(End Sepsis, 2021)

1) **Conclusion:** *Geospatial analysis reveals not just where disparities exist- but where change is needed the most*

Combining geospatial mapping with rate analysis improves identification of high-risk areas

2)**Recommendations:** Healthcare officials should use this method when implementing health policy changes and educational resources to prioritize allocated funds for training and resources for medical staff in identified "Hotspots" to bridge health disparities

For more accurate analysis clinical data collection should aim to minimize missing values, data regulatory oversight to limit incomplete data to ensure best practices in efforts to accurately display findings for better outcomes

3) **Future work:** Test and Evaluate changes/ policy/resources implemented with

Spatio-Temporal Modeling: Expanding the dataset beyond 2016–2018 would allow for **temporal mining** to see if specific policy changes are effectively reducing sepsis rates over time

Moraga, P. (2020). *Geospatial health data : modeling and visualization with R-INLA and Shiny*. Crc Press, Taylor & Francis Group.

References

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Thank you

Carmen Soto-MSDS Student

Email: carmensoto@arizona.edu

GitHub repo: carmssoto/INFO523_Final_Project_CarmenSoto: Carmen Soto Final Project